

Impact of Crude Oil Price Fluctuations on Regional Fuel Demand and Revenue Analysis (2018–2025)

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Subject : Visualization and storytelling with data

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Title: Crude Oil Market Analysis Report (2018–2025)

Subtitle: An Evaluation of Fuel Demand, Revenue Metrics, and Macroeconomic Influences across Indian Metro Cities

1.Introduction

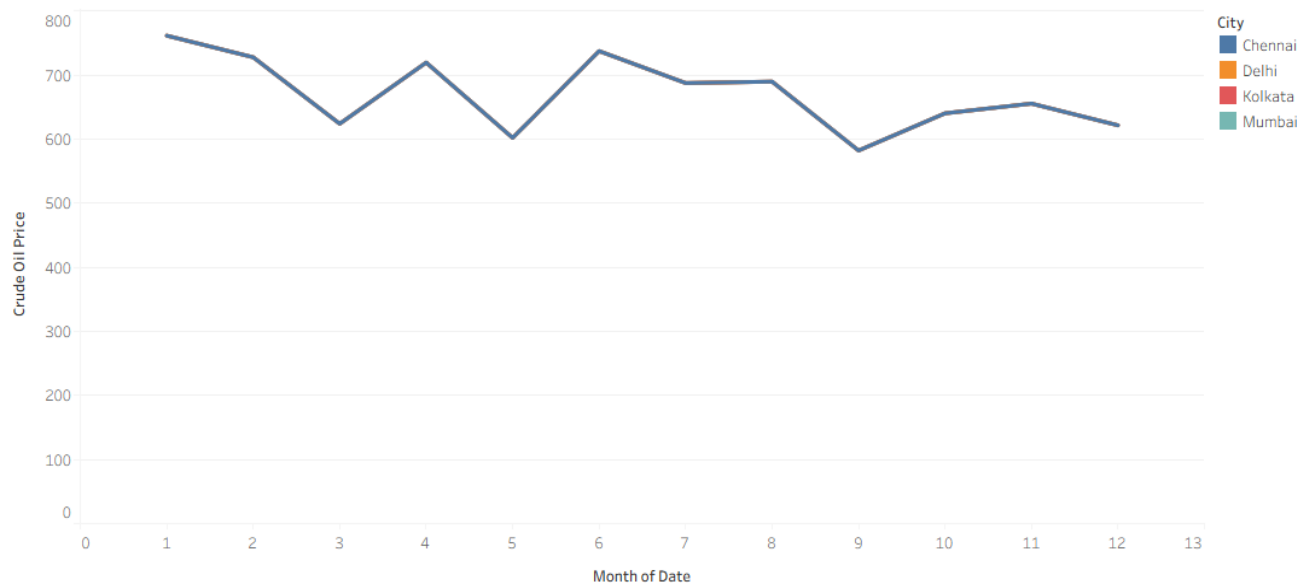
In rapid-growth economic landscapes like India, energy consumption patterns act as baseline indicator of regional economic health. This market analysis uses data visualization techniques to evaluate fuel market dynamics across the cities of Chennai, Delhi, Kolkata, and Mumbai from 2018 to 2025.

As global and domestic crude oil costs undergo unexpected shifts, municipal economies must navigate varying financial margins and localized demand elasticity. This report structures those market forces by analyzing key performance metrics—such as sales revenues, volume requirements, retail petrol pricing, and inflation indicators—allowing us to benchmark regional resilience and pinpoint the core macroeconomic drivers behind changing consumer demand.

2. Core Market Trends and Revenue Dynamics

A. Market Pricing Volatility

Crude Oil Price Fluctuation Trend (2018–2025)

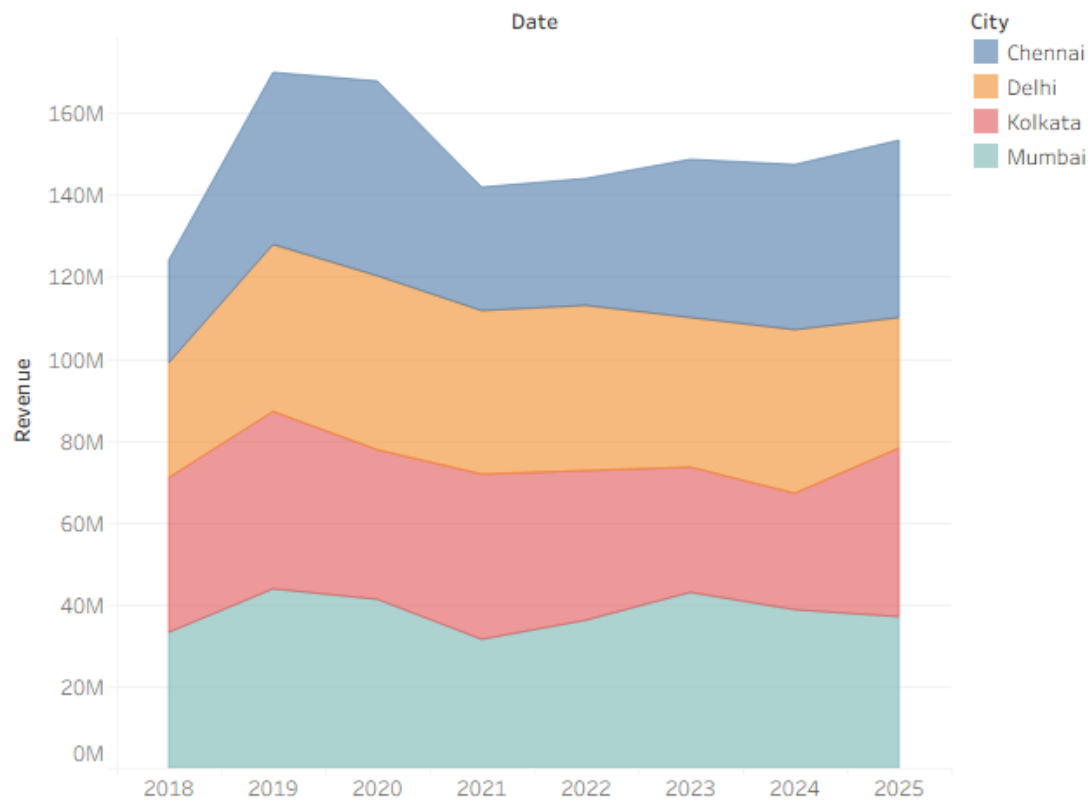


The trend of sum of Crude Oil Price for Date Month. Color shows details about City. The data is filtered on Month and Year. The Month filter keeps 12 of 12 members. The Year filter ranges from 2018 to 2025. The view is filtered on City, which keeps Chennai, Delhi, Kolkata and Mumbai.

This line visualization maps out the multi-year macro timeline for underlying raw material pricing. The graph reveals how core global fluctuations directly affect localized cost footprints across the regions. Observing the spikes and valleys provides critical context on supply chain impacts and shifting structural pricing mechanisms over time

B. Regional Progression Performance

Regional Revenue Growth



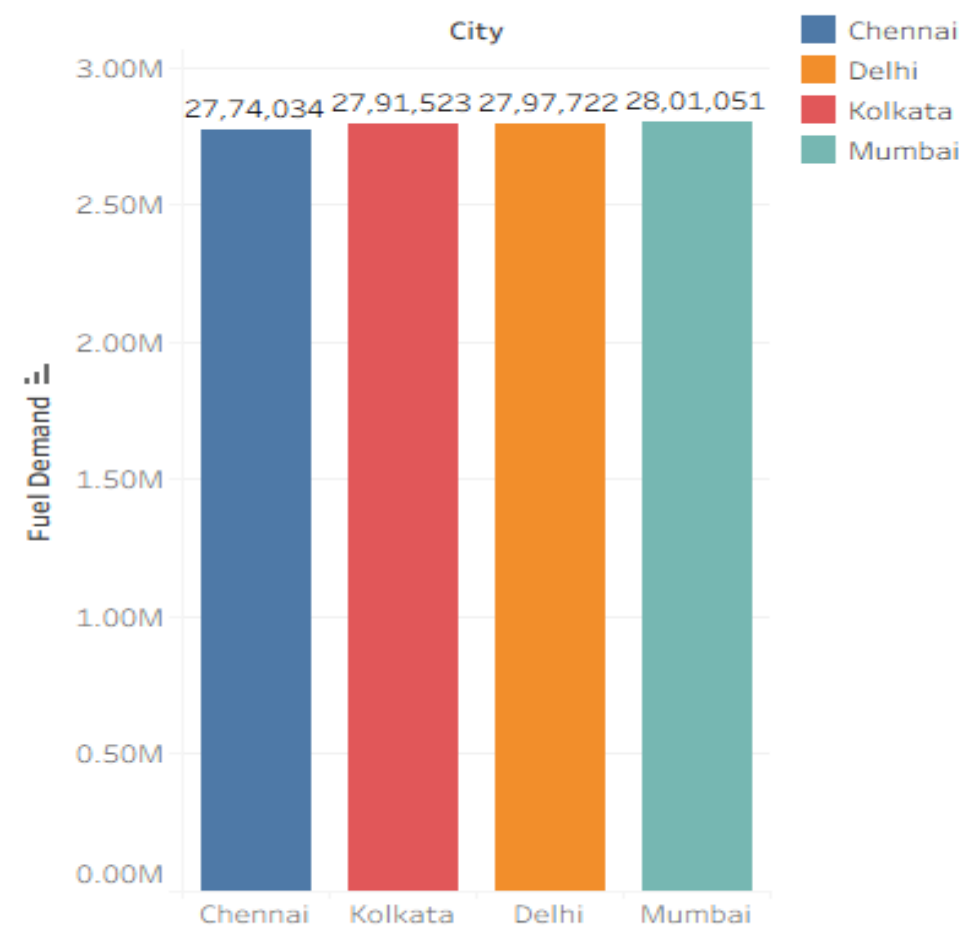
Sum of Revenue for each Date Year. Color shows details about City.

Utilizing an area chart approach, this visual layer highlights cumulative revenue generation grouped by the core urban markets over sequential years. By visualizing structural shifts over time, it highlights which municipal markets are accelerating their economic contribution versus those demonstrating stagnant scale patterns

3. Consumption Footprint and Geographic Distribution

A. Volumetric Comparison

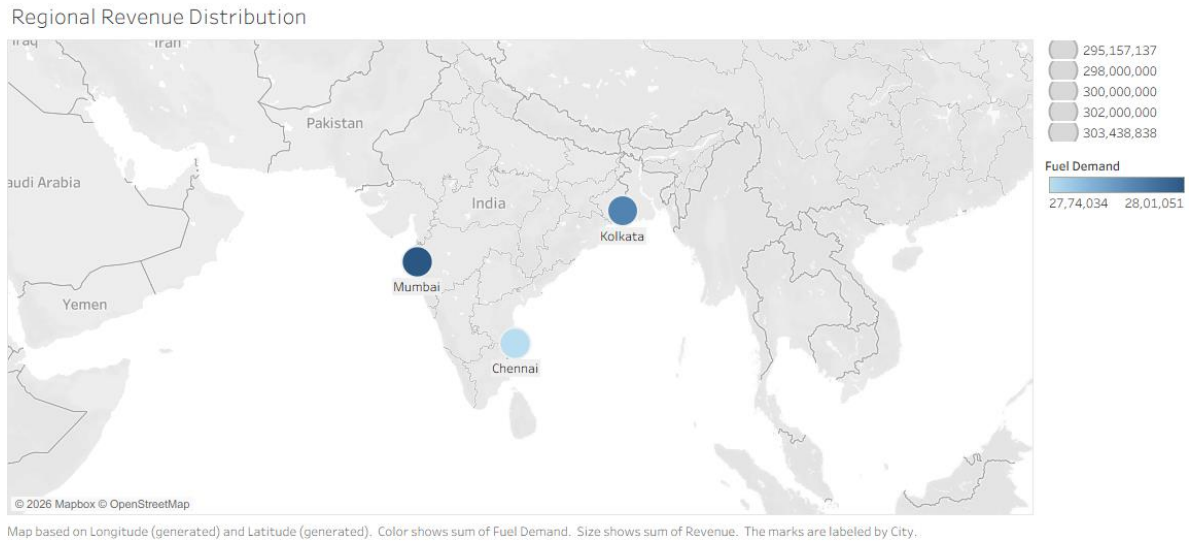
Fuel Demand Distribution by City



Sum of Fuel Demand for each City. Color shows details about City. The view is filtered on City, which keeps Chennai, Delhi, Kolkata and Mumbai.

The bar chart orders the city clusters by total accumulated structural requirements. This comparative visualization makes it instantly clear which specific urban industrial bases maintain the heaviest baseline energy requirements, allowing for targeted supply-side planning and distribution optimizations.

B. Geographic Mapping

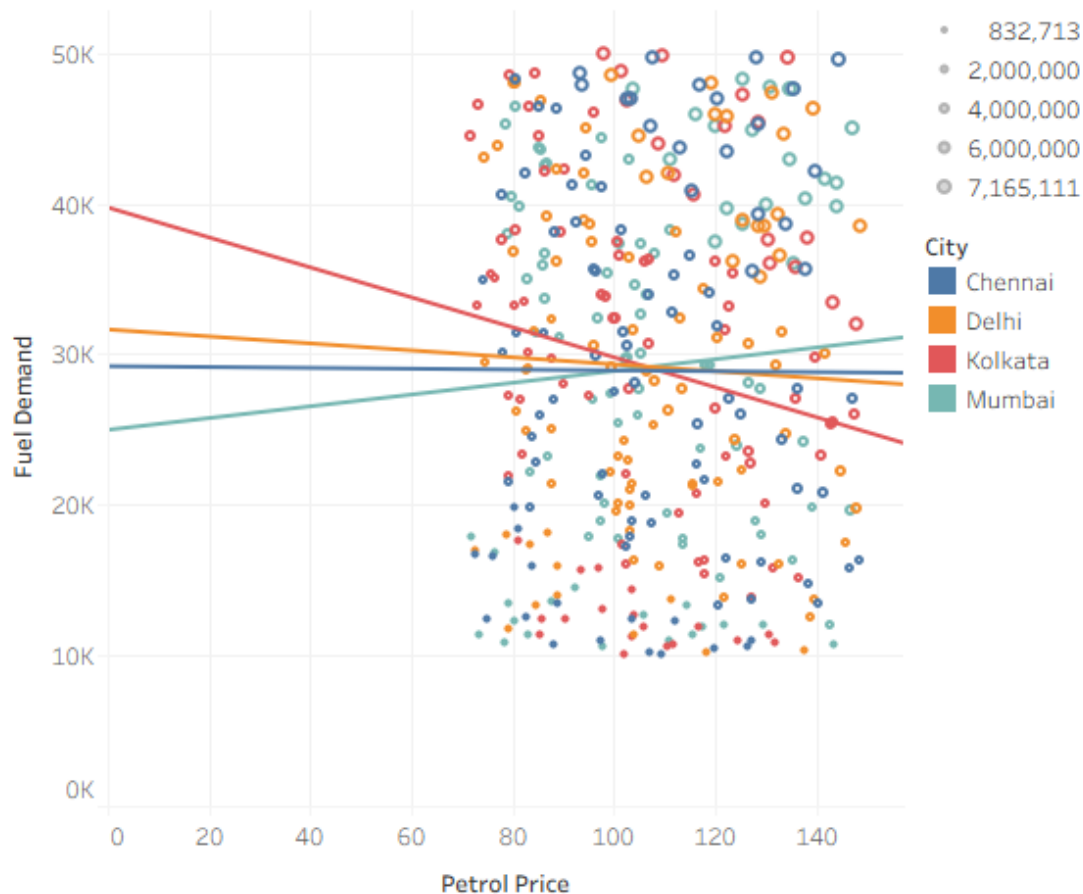


By plotting performance metrics directly onto a spatial geographic projection map, we bridge the gap between abstract numbers and true geographic locations. Variations in bubble scale and tint immediately highlight regional performance hubs, showcasing where logistical focus should be deployed to support high-yielding demand networks

4. Analytical Correlation and Macroeconomic Drivers

A. Price Elasticity Testing

Petrol Price vs Fuel Demand

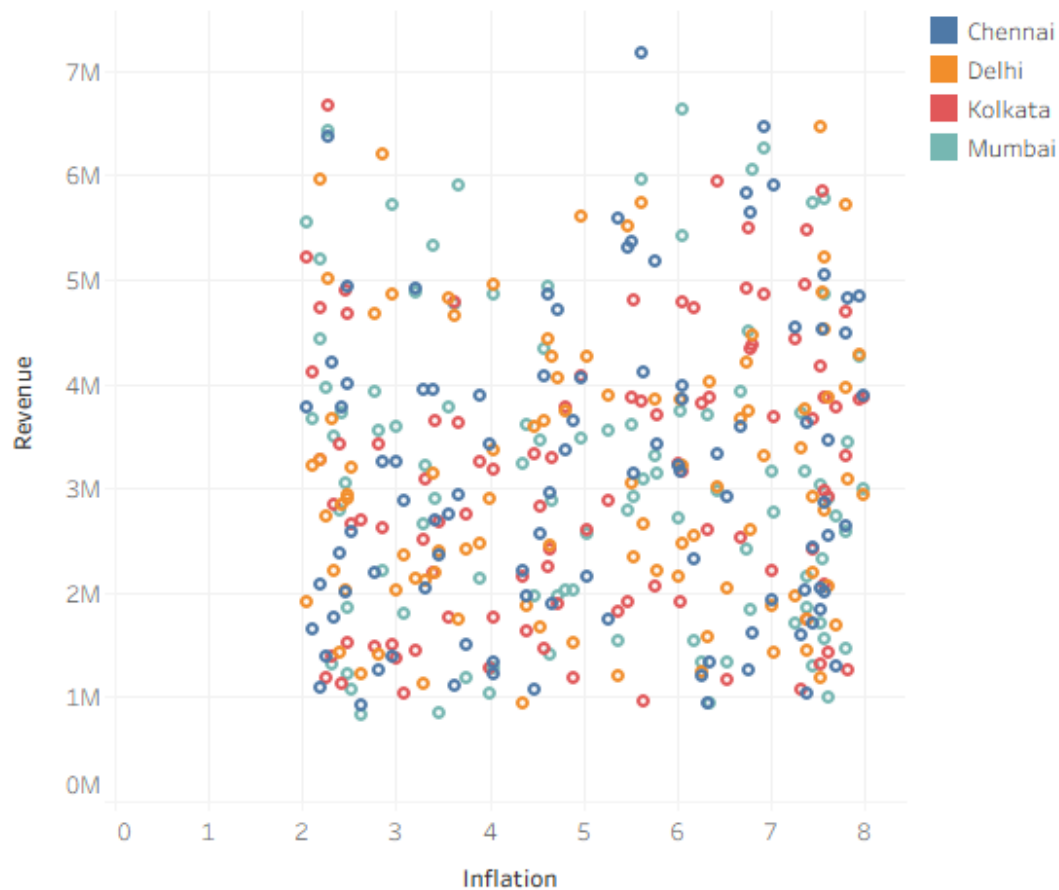


Petrol Price vs. Fuel Demand. Color shows details about City. Size shows Revenue. Details are shown for Year of Date.

This scatter visualization runs cross-variable checking between localized consumer retail costs and final volumetric purchase choices, backed by a linear trendline calculation. The slope direction acts as a direct metric for consumption elasticity, showing how resilient demand remains even when localized pricing moves higher.

B. Inflationary Pressure Verification

Inflation Impact on Revenue



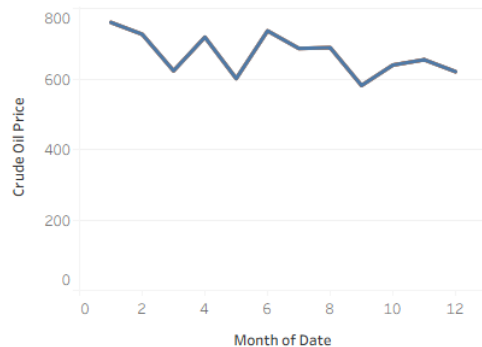
Inflation vs. Revenue. Color shows details about City. Details are shown for Year of Date and Day of Date.

This visualization tracks how underlying inflationary pressures correlate with overall nominal financial yields. This assessment is vital for distinguishing between organic volume-driven demand growth and artificial, price-inflated revenue increases caused by broader macroeconomic environment factors.

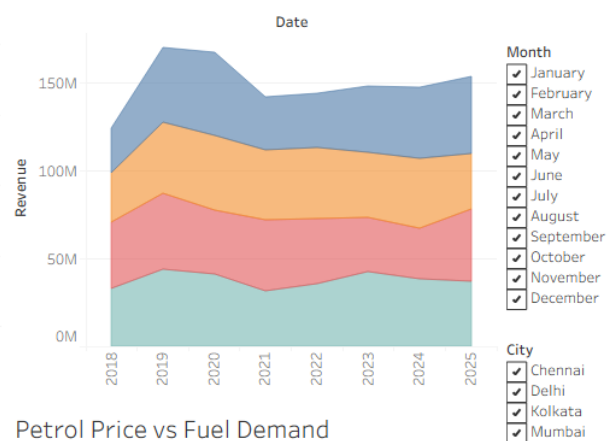
5. Strategic Conclusions & Recommendations

Crude Oil Market Analysis: Fuel Demand, Revenue and Regional Impact (2018–2025)

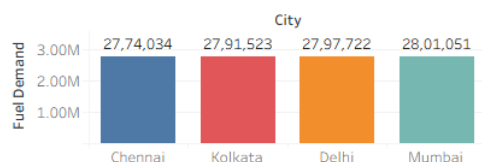
Crude Oil Price Fluctuation Trend (2018–2025)



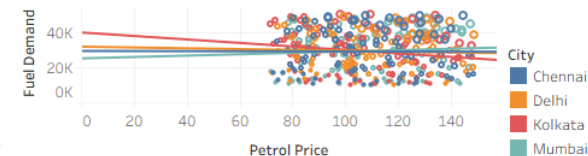
Regional Revenue Growth



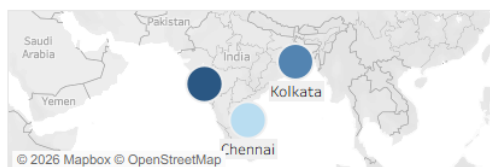
Fuel Demand Distribution by City



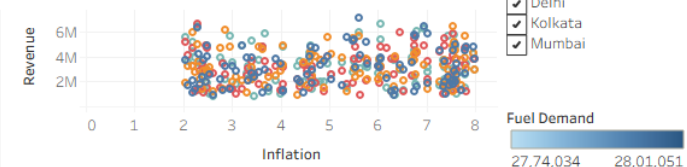
Petrol Price vs Fuel Demand



Regional Revenue Distribution



Inflation Impact on Revenue



- Market Resilience:** Despite significant crude oil price volatility over the multi-year timeline, major metro clusters—particularly high-demand hubs like Mumbai and Delhi—demonstrate relatively inelastic fuel demand, maintaining baseline volume requirements even during retail price peaks.
- Revenue Separation:** The correlation tracking between inflation and financial yields highlights that recent surges in nominal revenue are heavily driven by macroeconomic inflationary pressures rather than a true structural expansion in organic volumetric consumption.
- Strategic Takeaway:** Combining spatial geographic mapping with localized price elasticity testing underscores the necessity for regionalized distribution strategies. Supply chains should prioritize high-volume, price-resilient metropolitan zones to optimize logistical margins during future global crude oil price shocks.

